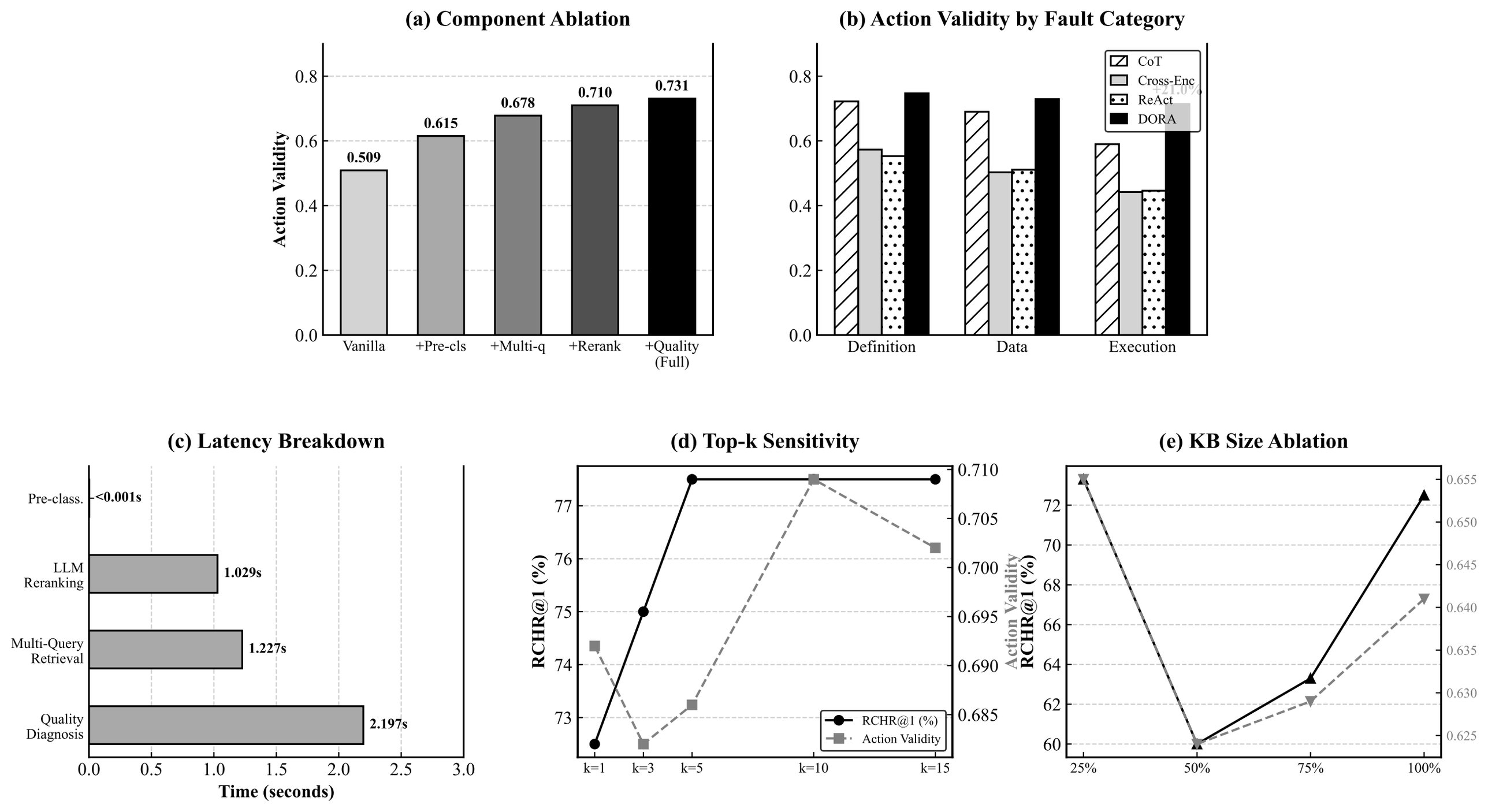

(a) Component Ablation
Action Validity
0.0
0.2
0.4
0.6
0.8
0.509
0.615
0.678
0.710
0.731
Vanilla
+Pre-cls
+Multi-q
+Rerank
+Quality (Full)
(b) Action Validity by Fault Category
CoT
Cross-Enc
ReAct
DORA
+21.0%
Definition
Data
Execution
(c) Latency Breakdown
Pre-class.
<0.001s
LLM Reranking
1.029s
Multi-Query Retrieval
1.227s
Quality Diagnosis
2.197s
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Time (seconds)
(d) Top-k Sensitivity
RCHR@1 (%)
Action Validity
73
74
75
76
77
0.685
0.690
0.695
0.700
0.705
0.710
k=1
k=3
k=5
k=10
k=15
RCHR@1 (%)
Action Validity
(e) KB Size Ablation
RCHR@1 (%)
Action Validity
60
62
64
66
68
70
72
0.625
0.630
0.635
0.640
0.645
0.650
0.655
25%
50%
75%
100%